

Export appliance

We can export a VM as an appliance in the Open Virtualisation Format (OVF). It comes in a two-file format.

- 1) **OVF file format:** In this format, several VM related files will be generated. For instance, there will be separate files for virtual hard disks, configurations and so on.
- 2) **OVA file format:** In this format, all VM related files will be archived into a single file and the .ova extension will be assigned to it.

By leveraging this feature, we can create a Golden Image of a VM and deploy multiple instances of it. OVF is a platform-independent, efficient, extensible, and open packaging and distribution format for VMs. As it is platform-independent, it allows the import of OVF virtual machines exported from VirtualBox into VMware Workstation Player and vice versa.

To export a VM, perform the steps listed below:

- 1) Select a VM from the VirtualBox manager. Navigate to the 'File->Export Appliance' option.
- 2) Select the VMs to be exported, and click the 'Next' button.
- 3) Provide the directory's location and OVF format version.
- 4) Provide the appliance settings and click the 'Export' button.

Import appliance

To import a VM, perform the steps given below:

- 1) Open VirtualBox Manager and navigate to the 'File->Import Appliance' option.
- 2) Select 'Virtual appliance' and click on the 'Next' button.
- 3) Verify appliance settings and click on the 'Import' button. You will see that a new VM appears in VirtualBox Manager's left pane.

Cloning a VM

VirtualBox also provides an option to clone existing VMs.

As the name suggests, it creates an exact copy of the VM. It supports the following two types of clones.

- 1) **Full clone:** In case of a full clone, it will duplicate all the VM's files. As this is a totally separate VM copy, we can easily move this VM to another host.
- 2) **Linked clone:** In case of linked clones, it will not copy virtual hard disks but, instead, it will take a snapshot of the original VM. It will create a new VM, but this one will refer to the virtual hard disks of the original VM. This is a space efficient clone operation, but the downside is that you cannot move the VM to another host as the original VM and the cloned one share the same virtual hard disks. To create a clone, perform the steps given below:

- 1) Select the VM from VirtualBox Manager. Right-click the VM and select the 'Clone' option.
- 2) Provide the name of the clone VM and click the 'Next' button.
- 3) Select the clone type and click the 'Clone' button. You will see that a new cloned VM appears in VirtualBox Manager's left pane.

Group VMs

VirtualBox allows you to create groups of VMs, and to manage and perform actions on them as a single entity. You can perform various actions on them like expanding/shrinking the group, renaming the group, or Start, Stop, Reset and Pause actions on the group of VMs.

To create a VM group, perform the following steps:

- 1) Select multiple VMs from VirtualBox Manager. Hold the 'Ctrl' key for multiple selections.
- 2) Right-click it and select the 'Group' option. This will create a VM group called 'New group' as shown in Figure 5.

If you right-click on the group, it will show various options like 'Add VM to group', 'Rename group', 'Ungroup', 'Start' and so on. To remove a VM from the group, just drag and drop that particular VM outside the group.

With snapshots, you can save a particular state of a VM for later use, at which point, you can revert to that state.

To take a snapshot, perform the following steps:

- 1) Select a VM from VirtualBox Manager.
- 2) Click the 'Machine Tools' drop down arrow from the toolbar and select the 'Snapshots' option.
- 3) Click the 'Take' button.
- 4) Enter the snapshot's name and description before clicking on the 'OK' button. Figure 6 depicts the above steps.

This window provides various options related to snapshots like Delete, Clone, Restore and so on. Click on the 'Properties' button to see more details about the selected snapshot.

Shared folders

Shared folders enable data sharing between the guest and host OS. They require VirtualBox guest additions to be installed inside the guest. This section describes the installation of guest additions along with the shared folder feature.

To enable the shared folder feature, perform the following steps:

- 1) Start the VM from VirtualBox Manager.
- 2) Go to the *Devices->Insert Guest Additions CD image* option. Follow the on-screen instructions to perform the guest additions installation. Figure 7 depicts the first two steps.
- 3) Navigate to *Devices->Shared Folders->Shared Folder Settings*.
- 4) Click the 'Add new shared folder' button. Enter the folder's name, its path and select 'Permissions'. Click the 'OK' button. Figure 8 illustrates the above steps.

You can mount the shared folder from the guest in the same way as an ordinary network share. Given below is the syntax for that:

```
mount -t vboxsf [-o OPTIONS] <sharename> <mountpoint>
```

Understanding virtual networking

This section delves deep into the aspects of VirtualBox's networking, which supports the network modes